KAKSHA PRACTICE PAPER · CLASS 9

Mathematics Mock Test 1

Time allowed: 3 hours **Maximum marks: 80**

Five sections A–E. Section A: 20 one-mark questions (MCQs and assertion–reason). Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks each. Section E: three case-study questions of 4 marks.

Section A

Multiple-choice questions (Q1–18) and assertion–reason questions (Q19–20). Choose the correct option.

1. Which of the following is an irrational number? **[1]**

(a) √4 (b) 0.333…
(c) 22/7 (d) √3

2. The value of (27)^(2/3) is **[1]**

(a) 6 (b) 9
(c) 18 (d) 3

3. On rationalising the denominator of 1/(√7 − √6), we get **[1]**

(a) √7 − √6 (b) √7 + √6
(c) (√7 + √6)/13 (d) 1/13

4. The degree of the polynomial $5x^3 + 4x^4 + 7$ is **[1]**

(a) 3 (b) 7
(c) 4 (d) 1

5. The zero of the polynomial p(x) = 2x − 5 is **[1]**

(a) −5/2 (b) 2/5
(c) 5 (d) 5/2

6. The remainder when $x^3 + 1$ is divided by x + 1 is **[1]**

(a) 0 (b) 1
(c) 2 (d) −2

7. Using a suitable identity, 103 × 107 equals **[1]**

(a) 11011 (b) 11021
(c) 11121 (d) 10921

8. The point (−3, 5) lies in the **[1]**

(a) first quadrant (b) second quadrant

(c) third quadrant (d) fourth quadrant

9. The line $2x + 3y = 12$ meets the y-axis at **[1]**

(a) (6, 0) (b) (0, 6)

(c) (0, 4) (d) (4, 0)

10. "The whole is greater than the part" is one of Euclid's **[1]**

(a) axioms (b) postulates

(c) definitions (d) theorems

11. Two angles form a linear pair. If one of them is 70°, the other is **[1]**

(a) 20° (b) 70°

(c) 90° (d) 110°

12. In $\triangle ABC$, $AB = AC$ and $\angle B = 50°$. Then $\angle A$ equals **[1]**

(a) 50° (b) 65°

(c) 80° (d) 100°

13. Which of the following is NOT a criterion for congruence of triangles? **[1]**

(a) SAS (b) ASA

(c) SSA (d) RHS

14. The angles of a quadrilateral are in the ratio 3 : 5 : 9 : 13. The largest angle is **[1]**

(a) 108° (b) 130°

(c) 144° (d) 156°

15. ABCD is a cyclic quadrilateral with $\angle A = 70°$. Then $\angle C$ equals **[1]**

(a) 70° (b) 110°

(c) 90° (d) 140°

16. Using Heron's formula, the area of a triangle with sides 3 cm, 4 cm and 5 cm is **[1]**

(a) 6 cm^2 (b) 12 cm^2

(c) 7.5 cm^2 (d) 10 cm^2

17. The volume of a sphere of radius 3 cm is **[1]**

(a) 12π cm^3 (b) 27π cm^3

(c) 36π cm^3 (d) 108π cm^3

18. The class mark of the class interval 10–20 is **[1]**

(a) 10 (b) 15

(c) 20 (d) 30

19. Assertion (A): 0.101001000100001... is an irrational number. Reason (R): Its decimal expansion is non-terminating and non-recurring. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

20. Assertion (A): $x = 2, y = 1$ is a solution of $2x + 3y = 7$. Reason (R): A linear equation in two variables has a unique solution. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Section B

Very short answer questions.

21. Find five rational numbers between 3/5 and 4/5. **[2]**

22. Factorise $12x^2 - 7x + 1$ by splitting the middle term. **[2]**

23. Two lines intersect at a point O. One of the angles formed is 40°. Find the other three angles, giving reasons. **[2]**

24. Find the curved surface area of a right circular cone whose slant height is 10 cm and base radius is 7 cm. ($\pi = 22/7$) **[2]**

25. Check which of the following are solutions of $x + 2y = 4$: (i) (0, 2) (ii) (2, 0) (iii) (4, 1). **[2]**

Section C

Short answer questions.

26. If $x = 3 + 2\sqrt{2}$, find the values of $x + 1/x$ and $x^2 + 1/x^2$. **[3]**

27. Factorise $x^3 - 23x^2 + 142x - 120$. **[3]**

28. Prove that the angles opposite to equal sides of an isosceles triangle are equal. **[3]**

29. ABCD is a rhombus and P, Q, R, S are the mid-points of AB, BC, CD and DA respectively. Show that PQRS is a rectangle. **[3]**

30. Find the area of a triangle whose sides are 13 cm, 14 cm and 15 cm using Heron's formula. **[3]**

31. A hemispherical bowl has an inner radius of 3.5 cm. How much water (in litres) can it hold? ($\pi = 22/7$) **[3]**

Section D

Long answer questions.

32. (i) Factorise $27x^3 + y^3 + z^3 - 9xyz$. (ii) Without actually calculating the cubes, find the value of $(-12)^3 + 7^3 + 5^3$. (iii) Expand $(2x - y + z)^2$. **[5]**

33. Prove that the sum of the angles of a triangle is 180°. In $\triangle ABC$, the angles A, B and C are in the ratio 2 : 3 : 4. Find all the angles and the exterior angle at C. **[5]**

34. Prove that equal chords of a circle subtend equal angles at the centre. Also, a chord of length 16 cm is drawn in a circle of radius 10 cm. Find the distance of the chord from the centre. **[5]**

35. A conical tent is 10 m high and the radius of its base is 24 m. Find (i) the slant height (ii) the cost of canvas required to make it at ₹70 per m^2 (iii) the volume of air inside the tent. ($\pi = 22/7$) **[5]**

Section E

Case-study questions (1 + 1 + 2 marks).

36. Case study — The school garden. A gardener marks four trees on a grid with the school gate as the origin: A(2, 3), B(−4, 2), C(−3, −5) and D(5, −1). (i) In which quadrant does tree C lie? (1) (ii) What is the abscissa of tree B? (1) (iii) A tap is on the x-axis, 3 units to the left of the gate. Write its coordinates. In which quadrant does tree D lie? (2) **[4]**

37. Case study — Taxi fare. In a city, the taxi fare is ₹8 for the first kilometre and ₹5 for each subsequent kilometre. Let x be the distance covered (in km) and y the total fare (in ₹). (i) Write a linear equation for this information. (1) (ii) Find the fare for a 10 km ride. (1) (iii) For what distance is the fare ₹78? Write any two points that lie on the graph of this equation. (2) **[4]**

38. Case study — Test marks. The marks (out of 50) of 40 students are: 0–10: 3, 10–20: 7, 20–30: 12, 30–40: 10, 40–50: 8. (i) Find the class mark of the class 20–30. (1) (ii) Which class has the maximum number of students? (1) (iii) How many students scored less than 30 marks? What percentage of students scored 40 or more? (2) **[4]**

Answer key and marking points

1. (d) $\sqrt{3}$

2. (b) 9

3. (b) $\sqrt{7} + \sqrt{6}$

4. (c) 4

5. (d) 5/2

6. (a) 0

7. (b) 11021

8. (b) second quadrant

9. (c) (0, 4)

10. (a) axioms

11. (d) 110°

12. (c) 80°

13. (c) SSA

14. (d) 156°

15. (b) 110°

16. (a) 6 cm²

17. (c) 36π cm³

18. (b) 15

19. (a) Both A and R are true, and R is the correct explanation of A

20. (c) A is true but R is false

21. Write both with denominator 30: $3/5 = 18/30$ and $4/5 = 24/30$. (1)
Five rational numbers: 19/30, 20/30, 21/30, 22/30, 23/30. (1)

22. Product $ac = 12$; split $-7x$ as $-4x - 3x$: $12x^2 - 4x - 3x + 1$. (1)
$= 4x(3x - 1) - 1(3x - 1) = (3x - 1)(4x - 1)$. (1)

23. The angle vertically opposite to 40° is 40° (vertically opposite angles are equal). (1)
The angles adjacent to it form linear pairs, so each is 180° − 40° = 140°. The angles are 40°, 140°, 40°, 140°. (1)

24. CSA = πrl. (1)
$= (22/7) \times 7 \times 10 = 220$ cm². (1)

25. (i) $0 + 2(2) = 4$ ✓ – a solution. (½)
(ii) $2 + 2(0) = 2 \neq 4$ – not a solution. (½)
(iii) $4 + 2(1) = 6 \neq 4$ – not a solution. (1)

26. $1/x = 1/(3 + 2\sqrt{2}) = (3 - 2\sqrt{2})/(9 - 8) = 3 - 2\sqrt{2}$. (1)
$x + 1/x = (3 + 2\sqrt{2}) + (3 - 2\sqrt{2}) = 6$. (1)
$x^2 + 1/x^2 = (x + 1/x)^2 - 2 = 36 - 2 = 34$. (1)

27. $p(1) = 1 - 23 + 142 - 120 = 0$, so $(x - 1)$ is a factor (Factor Theorem). (1)
Dividing, $x^3 - 23x^2 + 142x - 120 = (x - 1)(x^2 - 22x + 120)$. (1)
$x^2 - 22x + 120 = (x - 10)(x - 12)$. So the polynomial $= (x - 1)(x - 10)(x - 12)$. (1)

28. Given: $\triangle ABC$ with $AB = AC$. To prove: $\angle B = \angle C$. Construction: draw AD, the bisector of $\angle A$, meeting BC at D. (1)
In $\triangle BAD$ and $\triangle CAD$: $AB = AC$ (given), $\angle BAD = \angle CAD$ (construction), $AD = AD$ (common). So $\triangle BAD \cong \triangle CAD$ (SAS). (1½)
Hence $\angle ABD = \angle ACD$ (CPCT), i.e. $\angle B = \angle C$. (½)

29. In $\triangle ABC$, P and Q are mid-points of AB and BC, so $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$ (Mid-point Theorem). Similarly, in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$. So $PQ \parallel SR$ and $PQ = SR$, hence PQRS is a parallelogram. (1½)
Similarly, in $\triangle ABD$, $PS \parallel BD$. (½)
Diagonals of a rhombus are perpendicular, so $AC \perp BD$. Since $PQ \parallel AC$ and $PS \parallel BD$, $PQ \perp PS$, i.e. $\angle QPS = 90°$. A parallelogram with one right angle is a rectangle. (1)

30. $s = (13 + 14 + 15)/2 = 21$ cm. (1)
Area $= \sqrt{[s(s - a)(s - b)(s - c)]} = \sqrt{(21 \times 8 \times 7 \times 6)}$. (1)
$= \sqrt{7056} = 84$ cm². (1)

31. Volume $= (2/3)\pi r^3 = (2/3) \times (22/7) \times 3.5 \times 3.5 \times 3.5$. (1)
$= (2/3) \times 134.75 \approx 89.83$ cm³. (1½)
Since 1000 cm³ = 1 litre, it holds about 0.0898 litre. (½)

32. (i) Using $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ with $a = 3x$, $b = y$, $c = z$: (1)
$27x^3 + y^3 + z^3 - 9xyz = (3x + y + z)(9x^2 + y^2 + z^2 - 3xy - yz - 3zx)$. (1)
(ii) Since $-12 + 7 + 5 = 0$, $a^3 + b^3 + c^3 = 3abc$. (1)
Value $= 3 \times (-12) \times 7 \times 5 = -1260$. (½)
(iii) $(2x - y + z)^2 = 4x^2 + y^2 + z^2 - 4xy - 2yz + 4zx$. (1½)

33. Given $\triangle PQR$. Through P draw a line XPY parallel to QR. (1)
$\angle XPQ = \angle PQR$ and $\angle YPR = \angle PRQ$ (alternate interior angles). (1)
$\angle XPQ + \angle QPR + \angle YPR = 180°$ (angles on a straight line), so $\angle Q + \angle P + \angle R = 180°$. (1)
$2k + 3k + 4k = 180° \Rightarrow k = 20°$, so $\angle A = 40°$, $\angle B = 60°$, $\angle C = 80°$. (1)
Exterior angle at C $= 180° - 80° = 100°$ $(= \angle A + \angle B)$. (1)

34. Given: AB = CD are chords of a circle with centre O. To prove: $\angle AOB = \angle COD$. (½)
In $\triangle AOB$ and $\triangle COD$: $OA = OC$, $OB = OD$ (radii) and $AB = CD$ (given). So $\triangle AOB \cong \triangle COD$ (SSS). (1½)
Hence $\angle AOB = \angle COD$ (CPCT). (1)
The perpendicular from the centre bisects the chord, so half-chord = 8 cm. (1)
Distance $= \sqrt{(10^2 - 8^2)} = \sqrt{36} = 6$ cm. (1)

35. (i) $l = \sqrt{(h^2 + r^2)} = \sqrt{(100 + 576)} = \sqrt{676} = 26$ m. (1)
(ii) Canvas = CSA $= \pi rl = (22/7) \times 24 \times 26 = 13728/7$ m². (1)
Cost $= (13728/7) \times 70 =$ ₹1,37,280. (1)
(iii) Volume $= (1/3)\pi r^2 h = (1/3) \times (22/7) \times 576 \times 10 \approx 6034.29$ m³. (2)

36. (i) Both coordinates of C are negative, so it lies in quadrant III. (1)
(ii) Abscissa (x-coordinate) of B = −4. (1)
(iii) Tap = (−3, 0). (1) D(5, −1) has x positive, y negative, so it lies in quadrant IV. (1)

37. (i) $y = 8 + 5(x - 1)$, i.e. $y = 5x + 3$ (or $5x - y + 3 = 0$). (1)
(ii) $y = 5(10) + 3 =$ ₹53. (1)
(iii) $5x + 3 = 78 \Rightarrow x = 15$ km. (1) Points: (1, 8) and (2, 13) (any two correct points). (1)

38. (i) (20 + 30)/2 = 25. (1)
(ii) 20–30 (12 students). (1)
(iii) Less than 30: 3 + 7 + 12 = 22 students. (1) 40 or more: 8 students = (8/40) × 100 = 20%. (1)